Kristu Jayanti College
AUTONOMOUS Bengaluru
Reaccredited A++ Grade by NAAC | Affiliated to Bengaluru North University

SecureVote: Building a Robust Online Voting System with Python

Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Submitted By

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22BCAB37

Under the guidance of

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CERTIFICATE FROM MENTOR

This is to certify that this internship work titled "SecureVote: Building a Robust Online Voting System with Python" is based on an original project study conducted by **Mitanshu**, (**22BCAB37**) of V semester BCA B under my mentorship. This internship work has not formed the basis for the award of any degree/ diploma by Bangalore North University or any other university.

Place: Bengaluru

Date:

Name of the mentor

Dr. Hari Priya M P

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CERTIFICATE OF COMPLETION

This is to certify that the project entitled "**SecureVote: Building a Robust Online Voting System with Python"** has been satisfactorily completed by **Mitanshu**, **22BCAB37** in partial fulfillment of the award of the Bachelor of Computer Applications degree requirements prescribed by Kristu Jayanti College (Autonomous) Bengaluru (Affiliated to Bangalore North University) during the academic year **2023 -24**.

Internal Guide **Head of the Department**

Valued by Examiners

1: _____ **Centre: Kristu Jayanti College**

2: _____ **Date:**

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INTERNSHIP CERTIFICATE

INNOVATE

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that

MITANSHU

[AICTE STUDENT ID: **934904000000**]
COMPUTER SCIENCE,
KRISTU JAYANTI COLLEGE

has effectively completed a 8-week internship in **Python Programming & Development**, spanning from **11-Jun-2024** to **06-Aug-2024**. Additionally, the intern has satisfactorily completed and submitted a project titled "**SecureVote: Building a Robust Online Voting System with Python**". We extend our best wishes for the future endeavors.

Certificate No:
INNOVATE/ppd/000214

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DECLARATION

I, **Mitanshu** hereby declare that this internship work titled **SecureVote: Building a Robust Online Voting System with Python** is based on the original project conducted by me under the guidance of **Dr. Haripriya MP**

This has not been submitted earlier for the award of any other degree/diploma from Bangalore North University or any other University.

Signature
Mitanshu
22BCAB37

Bengaluru
Date:

ACKNOWLEDGEMENT

"The success is the result of preparation, hard work, and learning from failure."

It's my duty to acknowledge and thank the individuals who has contributed to the successful completion of the Internship.

I take this opportunity to express my profound and wholehearted thanks to **Rev. Fr. Dr. Augustine George, Principal** and **Rev. Fr.Dr. Lijo P Thomas, Vice Principal** and **Chief Financial Officer, Kristu Jayanti College, Bangalore.**

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The successful completion of my internship would not have been possible without my parent's

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Last but not the least I thank almighty God for giving me strength and good health throughout my internship and enabling me to complete it successfully.

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1. Introduction

1.1 Purpose

The "SecureVote" project addresses the increasing need for secure and transparent online voting systems by utilizing the Python programming language. With the rise of digital platforms, ensuring the integrity and anonymity of the voting process has become paramount. This project aims to create a comprehensive online voting platform that prioritizes security, usability, and verifiability. By incorporating advanced cryptographic techniques and secure protocols, SecureVote aspires to build a trusted system for both voters and election administrators. Python's versatility and reliability are leveraged to enhance the accessibility and inclusiveness of the democratic process in the digital era.

1.2 Goals

The primary goal of the SecureVote project is to develop a robust and trustworthy online voting system that allows voters to securely participate in elections from any location with internet access. The system aims to meet high security standards, ensuring the protection of voter data and the integrity of the voting process. By utilizing Python and its extensive libraries, the project seeks to create a scalable and customizable platform that can adapt to the evolving requirements of electoral processes worldwide. Through rigorous testing and adherence to industry standards, SecureVote aims to foster confidence in the fairness and security of online elections.

1.3 Overview of the Project

SecureVote focuses on building a secure and transparent online voting platform using Python. The project involves several key stages:

1. **Requirement Analysis**: Engaging with stakeholders to gather and analyze requirements, focusing on security, usability, and regulatory compliance.
2. **System Design**: Crafting the architecture and protocols, emphasizing cryptographic techniques for secure ballot casting, transmission, and tallying.

3. **Implementation**: Developing the system in Python, following best practices in software development and security.

4. **Cryptographic Integration**: Utilizing cryptographic libraries and algorithms for secure encryption, digital signatures, and verifiable computing.

5. **Testing and Validation**: Performing extensive testing to ensure security, reliability, and usability across various scenarios and threat models.

6. **Audit and Certification**: Engaging independent auditors for code reviews, penetration testing, and certification to meet industry standards.

The project uses Python, Django, and several Python libraries. The outcome is a scalable, secure, and user-friendly online voting system that enhances democratic participation.

2. Industry Profile

2.1 Overview of Innovate Intern:

The Innovate Intern Python Programming and Development internship offers a comprehensive, hands-on learning experience over eight weeks, designed to equip participants with essential Python skills and practical knowledge in software development. Interns begin with the fundamentals of Python, covering data types, control structures, functions, and modules. A significant focus is placed on cryptography and security, with interns working on projects like "SecureVote," an online voting platform that integrates cryptographic techniques and secure protocols to ensure data integrity and confidentiality. Additionally, interns gain experience in testing and validation, conducting unit and integration tests to ensure robust and reliable applications. The internship emphasizes following industry standards and best practices, preparing participants to tackle complex software projects and contribute effectively to the technology sector.

2.2 Company History

Innovate Intern is a dynamic and forward-thinking organization dedicated to providing high-quality internships and practical learning experiences in various fields of technology. Focused on bridging the gap between academic knowledge and industry demands, Innovate Intern offers meticulously designed programs that cater to the evolving needs of the tech industry. The company emphasizes hands-on training and real-world projects, ensuring that interns not only learn theoretical concepts but also apply them in practical scenarios. With a commitment to fostering innovation and excellence, Innovate Intern's programs cover a broad spectrum of topics, including Python programming, web development, cybersecurity, data science, and more.

Innovate Intern's internships are structured to provide a deep dive into specific technologies, supported by expert mentorship and a collaborative learning environment. Participants work on cutting-edge projects that mirror real industry challenges, equipping them with the skills and experience needed to excel in their careers. The company places a strong emphasis on following industry standards and best practices, ensuring that interns are well-versed in the latest developments and methodologies.

By partnering with educational institutions and industry leaders, Innovate Intern ensures that its programs are relevant, up-to-date, and aligned with the current market demands. The company's mission is to empower the next generation of tech professionals with the knowledge, skills, and confidence to drive innovation and make significant contributions to the field. Through its comprehensive and immersive internship programs, Innovate Intern is shaping the future of technology education and professional development.

2.3 Market Position and Trends

Innovate Intern has established a strong market position as a leading provider of high-quality, technology-focused internship programs. The company is recognized for its commitment to bridging the gap between academic learning and industry requirements, making it a preferred choice for students and aspiring tech professionals seeking practical, hands-on experience. Innovate Intern’s strategic partnerships with educational institutions and industry leaders enhance its credibility and reach, allowing it to attract a diverse pool of talented individuals. The company’s emphasis on real-world projects and adherence to industry standards sets it apart from competitors, ensuring that its programs remain relevant and valuable in the rapidly evolving tech landscape.

The market trend shows a growing demand for experiential learning opportunities, as employers increasingly seek candidates with practical skills and experience. Innovate Intern is well-positioned to capitalize on this trend, offering internships that are not only comprehensive and immersive but also aligned with the latest technological advancements and industry practices. The rise of remote work and online learning has further expanded the company’s reach, enabling it to offer flexible, accessible programs to a global audience. Innovate Intern’s focus on emerging technologies such as cybersecurity, data science, and AI ensures that its programs remain cutting-edge, preparing participants for the future demands of the tech industry. As the demand for skilled tech professionals continues to grow, Innovate Intern is poised for sustained growth and continued leadership in the field of technology education and professional development.

2.4 Technological Innovations

Innovate Intern is at the forefront of technological innovation, continually enhancing its internship programs with the latest advancements in the tech industry. The company leverages

cutting-edge technologies to create immersive and interactive learning experiences for its participants. By incorporating tools such as virtual labs, AI-driven learning platforms, and advanced coding environments, Innovate Intern ensures that interns receive a hands-on education that mirrors real-world scenarios. The integration of machine learning and data analytics into its curriculum allows for personalized learning paths, adapting to the unique needs and progress of each intern.

Innovate Intern's focus on emerging technologies, such as blockchain for secure voting systems, AI for intelligent automation, and cybersecurity protocols, prepares interns to tackle current and future technological challenges. The company's partnerships with tech giants and startups enable access to the latest software, tools, and industry insights, keeping its programs relevant and ahead of the curve. By fostering a culture of innovation and continuous improvement, Innovate Intern not only equips its interns with the necessary technical skills but also encourages creative problem-solving and critical thinking. This approach ensures that graduates are not just proficient in current technologies but are also adaptable and ready to lead in the ever-evolving tech landscape.

2.5 Technology and Framework

- Programming Languages:
  - Python
  - JavaScript

- Web Development Frameworks:
  - Django (for Python)
  - Flask (for Python)
  - React.js (for JavaScript)
  - Node.js (for JavaScript)

- Databases:
  - PostgreSQL
  - MySQL
  - MongoDB

- Cryptographic Libraries:
  - cryptography (Python)
  - PyCryptodome (Python)

- Web Development Tools:
  - HTML5
  - CSS3
  - Bootstrap

- APIs and RESTful Services:
  - Django REST framework
  - Express.js (Node.js)

- Version Control and Collaboration:
  - Git
  - GitHub
  - Cloud and Deployment:
  - AWS (Amazon Web Services)
  - Heroku
  - Docker

- Machine Learning and Data Science:
  - TensorFlow
  - Pandas
  - NumPy

- Testing Frameworks:
  - pytest (Python)
  - Selenium (for web application testing)

- -Security Standards:
  - NIST SP 800-53
  - E2E-VIV standards for electronic voting systems

3. Internship Work / Project Description

3.1 Training and Initial Tasks

The training and initial tasks at Innovate Intern in Python Programming and Development are meticulously designed to build a strong foundation and progressively advance interns' skills. The program begins with comprehensive training on Python fundamentals, including data types, control structures, functions, and modules, ensuring that interns grasp the core concepts of the language. Following this, interns are introduced to object-oriented programming (OOP) principles, which are essential for developing scalable and maintainable code.

To instill best practices in software development, interns are assigned tasks that involve writing unit tests, debugging code, and performing code reviews. They also engage in hands-on projects, such as developing a basic blog or a to-do list application, which reinforce their learning and provide practical experience. Throughout the training, interns receive continuous mentorship and feedback, ensuring they understand each concept thoroughly and are prepared to tackle more complex challenges as the program progresses. This structured approach ensures that by the end of the initial phase, interns are well-equipped with the essential skills needed for advanced Python programming and development tasks.

3.2 Project Overview

The "SecureVote: Building a Robust Online Voting System with Python" project aims to develop a secure and transparent online voting platform using the Python programming language. Addressing the critical need for secure electronic voting, the project focuses on ensuring data integrity, voter anonymity, and the overall integrity of the electoral process. SecureVote employs advanced cryptographic techniques and secure protocols to safeguard the voting process, from ballot casting to tallying. Extensive testing and validation processes are implemented to ensure the system's reliability and security across various scenarios. By adhering to industry standards and best practices, SecureVote aspires to build trust among voters and election administrators, offering a scalable, user-friendly solution for modern democratic participation.

3.3 Project Requirements and Planning

- **Project Requirements**
  - User Authentication: Secure login and registration.
  - Voting Process: Cast votes anonymously.
  - Vote Tallying: Count votes and generate results.
  - Results Display: Show real-time results.
  - Audit Trail: Log activities for transparency.
  - Security: Implement encryption and protect against tampering.
- **Project Planning**
  - Define Scope: Goals, deliverables, and constraints.
  - Requirement Analysis: Gather detailed requirements.
  - Design:
  - Architecture: Plan the overall structure.
  - UI/UX: Create basic interfaces.
  - Implementation:
  - Python Code: Develop the voting logic, authentication, and data handling.
- **Testing:**
- **Unit Testing: Test individual components.**
- **System Testing: Verify overall functionality.**
  - Deployment: Prepare and run the application.
  - Documentation: Document code and user guides.
  - Support: Provide basic maintenance and updates.

3.4 Key Responsibilities

My responsibilities included:

- Design and Implement Structure: Develop Python scripts to handle user authentication, voting logic, and result generation.
- Ensure Security: Implement encryption and other security measures to protect user data and voting integrity.
- Develop User Interface: Create a simple and intuitive command-line or basic GUI interface for user interactions.

- Populate Data: Integrate sample voting data and set up mock elections for testing purposes.
- Testing and Refinement: Test the system for functionality, security, and performance, and refine based on test results and feedback.

3.5 Development Process

SecureVote: Building a Robust Online Voting System with Python

1. Design and Implement Structure:

Create Python scripts to handle core functionalities such as user authentication, voting logic, and result tallying. Structure your code to include modules for different components like user management, vote processing, and results generation.

2. Ensure Security:

Implement security measures in your Python code to protect user data and voting integrity. This includes encryption for sensitive information, secure authentication methods, and measures to prevent tampering.

3. Develop User Interface:

Develop a basic command-line or simple GUI for user interactions. Ensure that the interface is intuitive and allows users to easily log in, cast their votes, and view results.

4. Populate Data:

Integrate sample voting data and create mock elections to test the system. Ensure that the data is varied and covers different scenarios to thoroughly evaluate system performance.

- **5. Testing and Refinement:**

Test the system for functionality, security, and performance. Verify that all features work as intended, and make necessary adjustments based on test results and feedback to improve the system's reliability and user experience.

3.6 Screenshots:

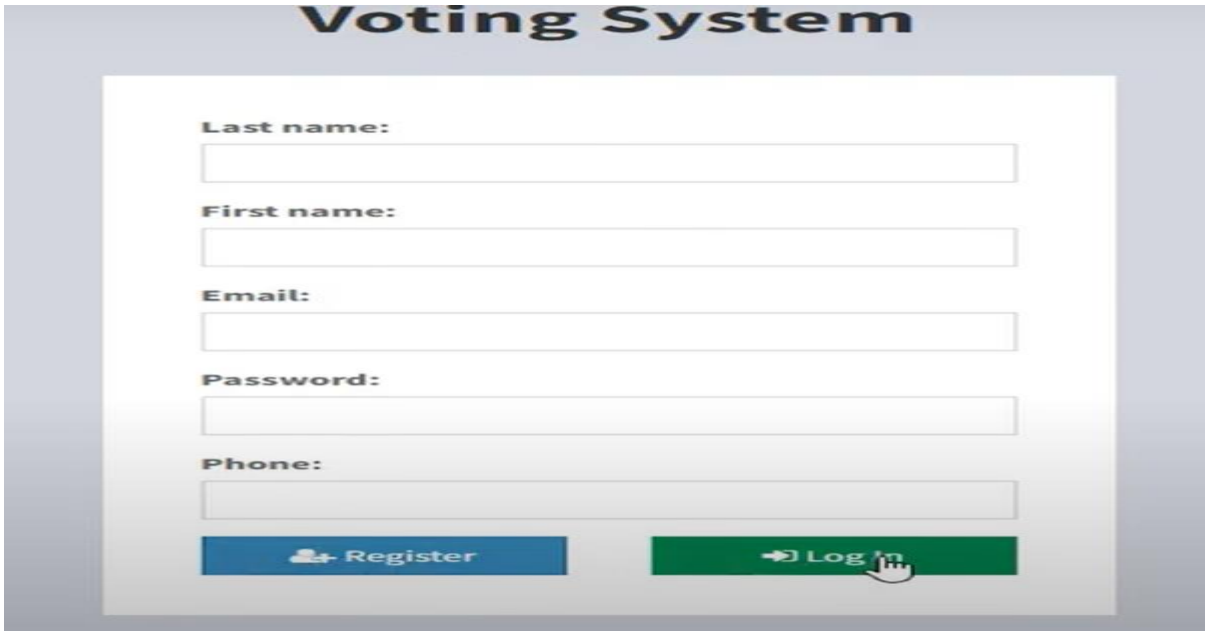

Fig 1: Voting Registration Form

Fig 2: Voter Login

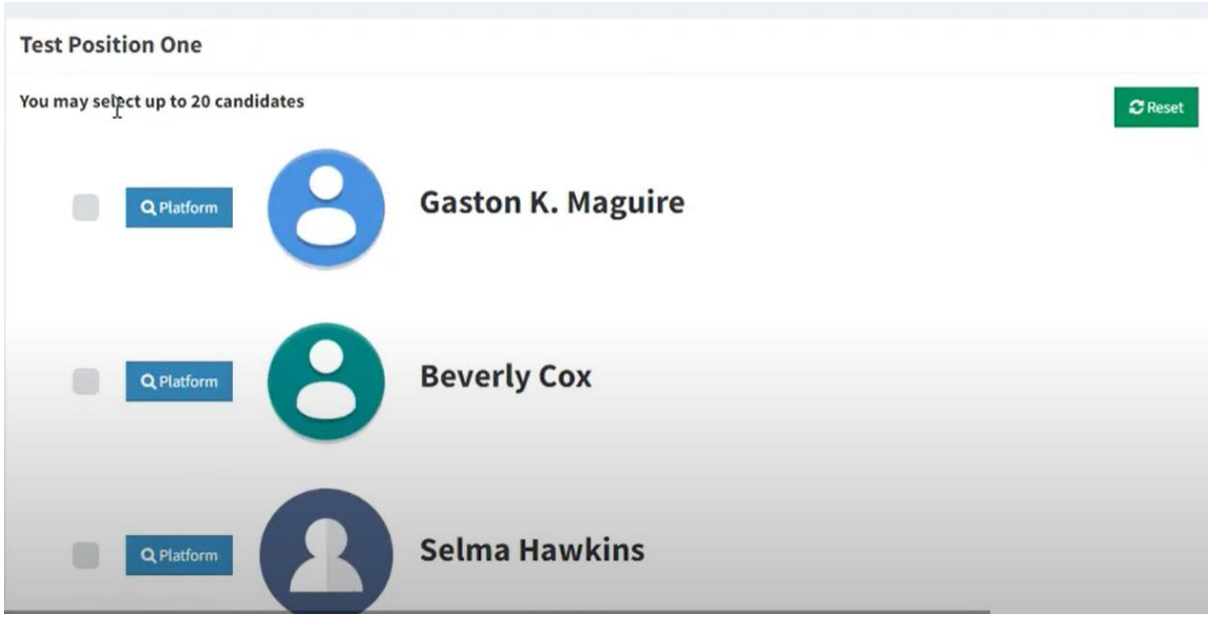

Fig 3: Test Position One Candidates

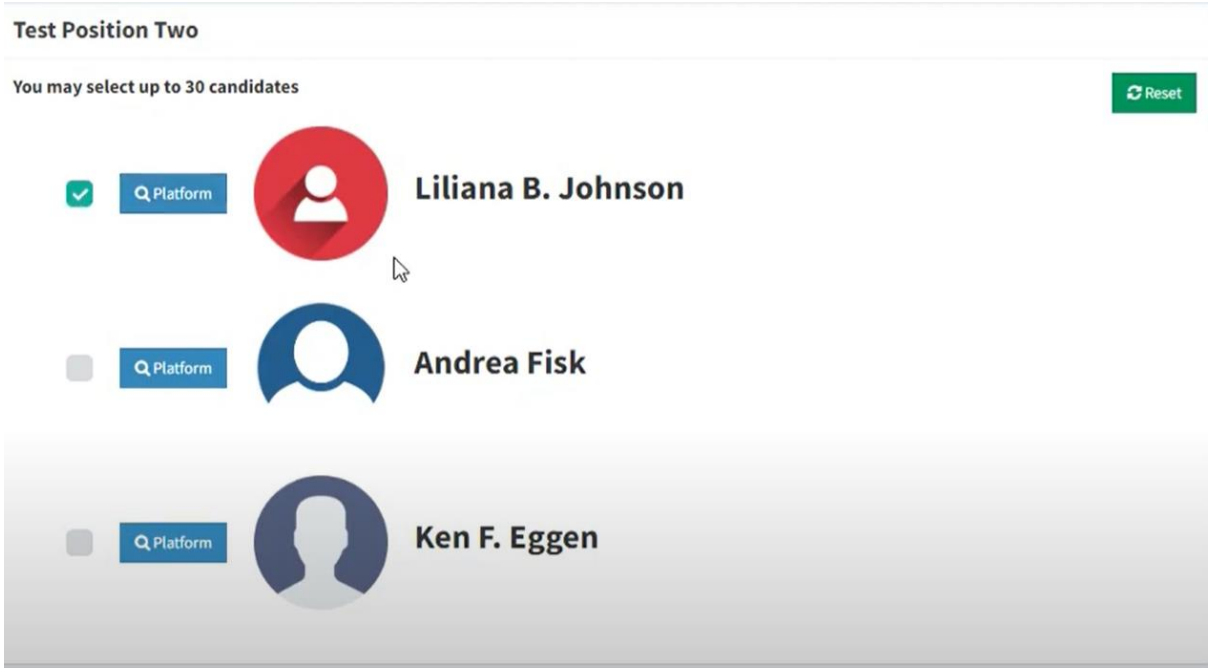

Fig 4: Test Position Two Candidates

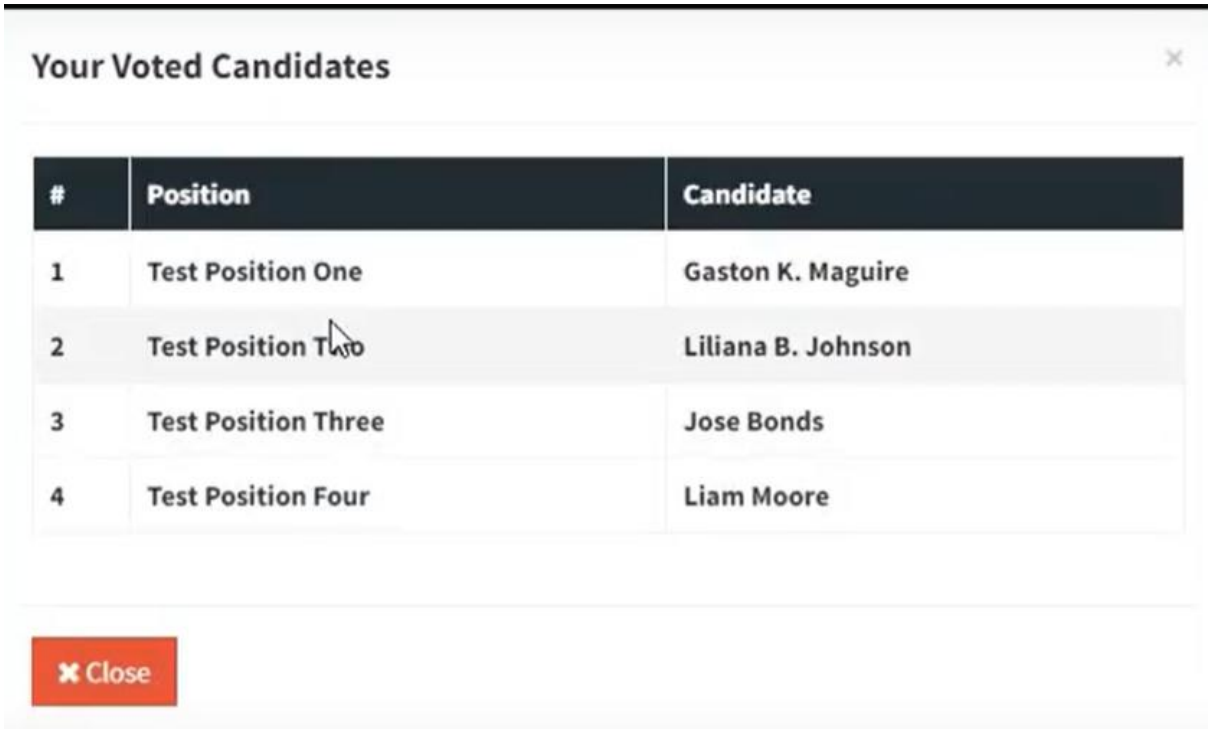

Fig 5: Voted Candidates Positioning

Fig 6: Admin Login

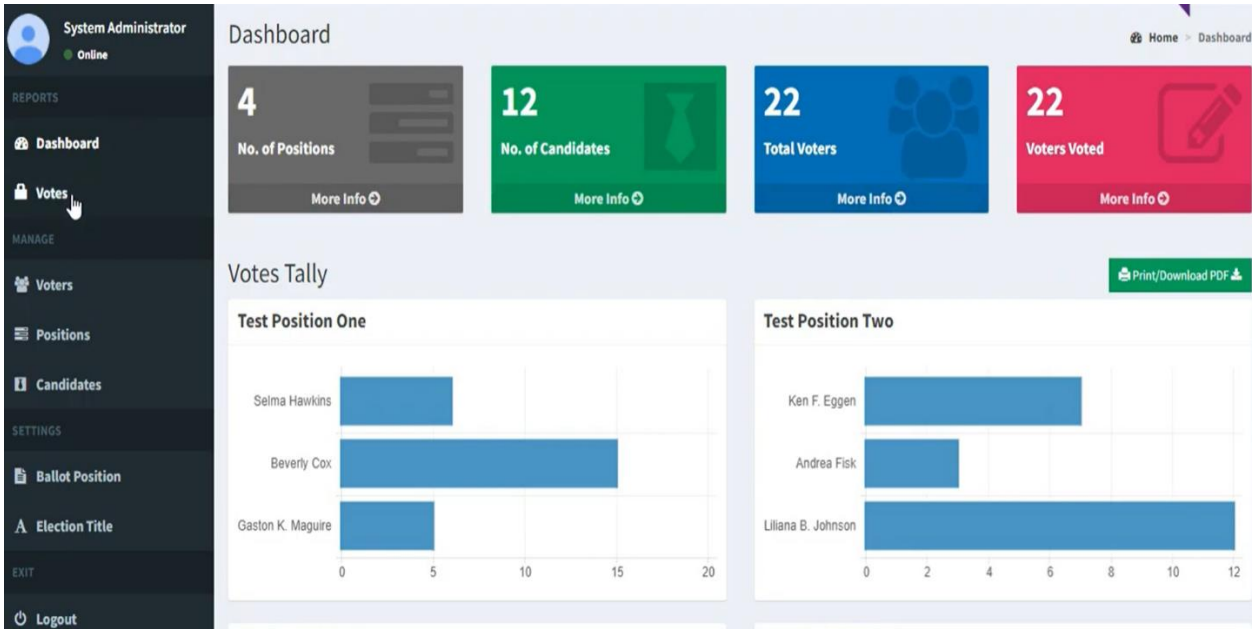

Fig 7: Admin Dashboard

3.7 Code

```
from django.shortcuts import render, reverse, redirect

from voting.models import Voter, Position, Candidate, Votes

from account.models import CustomUser

from account.forms import CustomUserForm

from voting.forms import *

from django.contrib import messages

from django.http import JsonResponse, HttpResponse

from django.conf import settings

import json  # Not used

from django_renderpdf.views import PDFView


def find_n_winners(data, n):

    """Read More

    https://www.geeksforgeeks.org/python-program-to-find-n-largest-elements-from-a-list/

    """
```

```
    final_list = []
    candidate_data = data[:]
    # print("Candidate = ", str(candidate_data))
    for i in range(0, n):
        max1 = 0
        if len(candidate_data) == 0:
            continue
        this_winner = max(candidate_data, key=lambda x: x['votes'])
        # TODO: Check if None
        this = this_winner['name'] + \
            " with " + str(this_winner['votes']) + " votes"
        final_list.append(this)
        candidate_data.remove(this_winner)
    return ",  ".join(final_list)

class PrintView(PDFView):
    template_name = 'admin/print.html'
    prompt_download = True

    @property
    def download_name(self):
        return "result.pdf"

    def get_context_data(self, *args, **kwargs):
        title = "E-voting"
        try:
            file = open(settings.ELECTION_TITLE_PATH, 'r')
```

```
        title = file.read()
    except:
        pass
    context = super().get_context_data(*args, **kwargs)
    position_data = {}
    for position in Position.objects.all():
        candidate_data = []
        winner = ""
        for candidate in Candidate.objects.filter(position=position):
            this_candidate_data = {}
            votes = Votes.objects.filter(candidate=candidate).count()
            this_candidate_data['name'] = candidate.fullname
            this_candidate_data['votes'] = votes
            candidate_data.append(this_candidate_data)
        print("Candidate Data For  ", str(
            position.name), " = ", str(candidate_data))
        # ! Check Winner
        if len(candidate_data) < 1:
            winner = "Position does not have candidates"
        else:
            # Check if max_vote is more than 1
            if position.max_vote > 1:
                winner = find_n_winners(candidate_data, position.max_vote)
            else:

                winner = max(candidate_data, key=lambda x: x['votes'])
                if winner['votes'] == 0:
```

```
                winner = "No one voted for this yet position, yet."
            else:
                """
                        https://stackoverflow.com/questions/18940540/how-can-i-count-the-occurrences-of-an-item-in-a-list-of-dictionaries
                """
                count = sum(1 for d in candidate_data if d.get(
                    'votes') == winner['votes'])
                if count > 1:
                    winner = f"There are {count} candidates with {winner['votes']} votes"
                else:
                    winner = "Winner : " + winner['name']
        print("Candidate Data For  ", str(
            position.name), " = ", str(candidate_data))
        position_data[position.name] = {
            'candidate_data': candidate_data, 'winner': winner, 'max_vote': position.max_vote}
    context['positions'] = position_data
    print(context)
    return context


def dashboard(request):
    positions = Position.objects.all().order_by('priority')
    candidates = Candidate.objects.all()
    voters = Voter.objects.all()
    voted_voters = Voter.objects.filter(voted=1)
    list_of_candidates = []
    votes_count = []
```

```
    chart_data = {}

    for position in positions:
        list_of_candidates = []
        votes_count = []
        for candidate in Candidate.objects.filter(position=position):
            list_of_candidates.append(candidate.fullname)
            votes = Votes.objects.filter(candidate=candidate).count()
            votes_count.append(votes)
        chart_data[position] = {
            'candidates': list_of_candidates,
            'votes': votes_count,
            'pos_id': position.id
        }

    context = {
        'position_count': positions.count(),
        'candidate_count': candidates.count(),
        'voters_count': voters.count(),
        'voted_voters_count': voted_voters.count(),
        'positions': positions,
        'chart_data': chart_data,
        'page_title': "Dashboard"
    }
    return render(request, "admin/home.html", context)

def voters(request):
```

```
    voters = Voter.objects.all()
    userForm = CustomUserForm(request.POST or None)
    voterForm = VoterForm(request.POST or None)
    context = {
        'form1': userForm,
        'form2': voterForm,
        'voters': voters,
        'page_title': 'Voters List'
    }
    if request.method == 'POST':
        if userForm.is_valid() and voterForm.is_valid():
            user = userForm.save(commit=False)
            voter = voterForm.save(commit=False)
            voter.admin = user
            user.save()
            voter.save()
            messages.success(request, "New voter created")
        else:
            messages.error(request, "Form validation failed")
    return render(request, "admin/voters.html", context)


def view_voter_by_id(request):
    voter_id = request.GET.get('id', None)
    voter = Voter.objects.filter(id=voter_id)
    context = {}
    if not voter.exists():
        context['code'] = 404
```

```
    else:
        context['code'] = 200
        voter = voter[0]
        context['first_name'] = voter.admin.first_name
        context['last_name'] = voter.admin.last_name
        context['phone'] = voter.phone
        context['id'] = voter.id
        context['email'] = voter.admin.email
    return JsonResponse(context)

def view_position_by_id(request):
    pos_id = request.GET.get('id', None)
    pos = Position.objects.filter(id=pos_id)
    context = {}
    if not pos.exists():
        context['code'] = 404
    else:
        context['code'] = 200
        pos = pos[0]
        context['name'] = pos.name
        context['max_vote'] = pos.max_vote
        context['id'] = pos.id
    return JsonResponse(context)

def updateVoter(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
```

```
    try:
        instance = Voter.objects.get(id=request.POST.get('id'))
        user = CustomUserForm(request.POST or None, instance=instance.admin)
        voter = VoterForm(request.POST or None, instance=instance)
        user.save()
        voter.save()
        messages.success(request, "Voter's bio updated")
    except:
        messages.error(request, "Access To This Resource Denied")

    return redirect(reverse('adminViewVoters'))

def deleteVoter(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
    try:
        admin = Voter.objects.get(id=request.POST.get('id')).admin
        admin.delete()
        messages.success(request, "Voter Has Been Deleted")
    except:
        messages.error(request, "Access To This Resource Denied")

    return redirect(reverse('adminViewVoters'))

def viewPositions(request):
    positions = Position.objects.order_by('-priority').all()
    form = PositionForm(request.POST or None)
```

```
    context = {
        'positions': positions,
        'form1': form,
        'page_title': "Positions"
    }
    if request.method == 'POST':
        if form.is_valid():
            form = form.save(commit=False)
            form.priority = positions.count() + 1  # Just in case it is empty.
            form.save()
            messages.success(request, "New Position Created")
        else:
            messages.error(request, "Form errors")
    return render(request, "admin/positions.html", context)


def updatePosition(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
    try:
        instance = Position.objects.get(id=request.POST.get('id'))
        pos = PositionForm(request.POST or None, instance=instance)
        pos.save()
        messages.success(request, "Position has been updated")
    except:
        messages.error(request, "Access To This Resource Denied")


    return redirect(reverse('viewPositions'))
```

```
def deletePosition(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
    try:
        pos = Position.objects.get(id=request.POST.get('id'))
        pos.delete()
        messages.success(request, "Position Has Been Deleted")
    except:
        messages.error(request, "Access To This Resource Denied")

    return redirect(reverse('viewPositions'))

def viewCandidates(request):
    candidates = Candidate.objects.all()
    form = CandidateForm(request.POST or None, request.FILES or None)
    context = {
        'candidates': candidates,
        'form1': form,
        'page_title': 'Candidates'
    }
    if request.method == 'POST':
        if form.is_valid():
            form = form.save()
            messages.success(request, "New Candidate Created")
        else:
            messages.error(request, "Form errors")
```

```
    return render(request, "admin/candidates.html", context)


def updateCandidate(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
    try:
        candidate_id = request.POST.get('id')
        candidate = Candidate.objects.get(id=candidate_id)
        form = CandidateForm(request.POST or None,
                             request.FILES or None, instance=candidate)
        if form.is_valid():
            form.save()
            messages.success(request, "Candidate Data Updated")
        else:
            messages.error(request, "Form has errors")
    except:
        messages.error(request, "Access To This Resource Denied")


    return redirect(reverse('viewCandidates'))


def deleteCandidate(request):
    if request.method != 'POST':
        messages.error(request, "Access Denied")
    try:
        pos = Candidate.objects.get(id=request.POST.get('id'))
        pos.delete()
        messages.success(request, "Candidate Has Been Deleted")
```

```
    except:
        messages.error(request, "Access To This Resource Denied")

    return redirect(reverse('viewCandidates'))

def view_candidate_by_id(request):
    candidate_id = request.GET.get('id', None)
    candidate = Candidate.objects.filter(id=candidate_id)
    context = {}
    if not candidate.exists():
        context['code'] = 404
    else:
        candidate = candidate[0]
        context['code'] = 200
        context['fullname'] = candidate.fullname
        previous = CandidateForm(instance=candidate)
        context['form'] = str(previous.as_p())
    return JsonResponse(context)

def ballot_position(request):
    context = {
        'page_title': "Ballot Position"
    }
    return render(request, "admin/ballot_position.html", context)

def update_ballot_position(request, position_id, up_or_down):
    try:
```

```
    context = {
        'error': False
    }
    position = Position.objects.get(id=position_id)
    if up_or_down == 'up':
        priority = position.priority - 1
        if priority == 0:
            context['error'] = True
            output = "This position is already at the top"
        else:
            Position.objects.filter(priority=priority).update(
                priority=(priority+1))
            position.priority = priority
            position.save()
            output = "Moved Up"
    else:
        priority = position.priority + 1
        if priority > Position.objects.all().count():
            output = "This position is already at the bottom"
            context['error'] = True
        else:
            Position.objects.filter(priority=priority).update(
                priority=(priority-1))
            position.priority = priority
            position.save()
            output = "Moved Down"
    context['message'] = output
```

```
    except Exception as e:
        context['message'] = e

    return JsonResponse(context)

def ballot_title(request):
    from urllib.parse import urlparse
    url = urlparse(request.META['HTTP_REFERER']).path
    from django.urls import resolve
    try:
        redirect_url = resolve(url)
        title = request.POST.get('title', 'No Name')
        file = open(settings.ELECTION_TITLE_PATH, 'w')
        file.write(title)
        file.close()
        messages.success(
            request, "Election title has been changed to " + str(title))
        return redirect(url)
    except Exception as e:
        messages.error(request, e)
        return redirect("/")

def viewVotes(request):
    votes = Votes.objects.all()
    context = {
        'votes': votes,
        'page_title': 'Votes'
```

```
    }
    return render(request, "admin/votes.html", context)


def resetVote(request):
    Votes.objects.all().delete()
    Voter.objects.all().update(voted=False, verified=False, otp=None)
    messages.success(request, "All votes has been reset")
    return redirect(reverse('viewVotes'))
```

3.8 Tools and Technologies Used

- **Python:**
- **Version:** Ensure you have Python 3.x installed. The code is compatible with Python 3.
- **Installation:** You can download it from the official Python website.
- **Text Editor or IDE:**
- **Visual Studio Code, PyCharm, or any text editor:** Use an editor or integrated development environment (IDE) to write and edit the code.
- **Installation:** Download from their respective websites (VS Code, PyCharm).
- **Command Line Interface (CLI):**
- **Terminal (macOS/Linux) or Command Prompt (Windows):** Use the command line to run the Python script.
- **Usage:** Navigate to the directory where your Python file is saved and run the script with python filename.py.
- **Hashing Library (built-in):**
- **hashlib:** This Python standard library is used for hashing passwords.
- **Installation:** No additional installation is required as it is part of Python's standard library.

3.9 Project Phases

The project was divided into several phases:

1. **Planning and Requirements Gathering:**

Define the system's goals, such as secure user authentication, voting functionality, and result generation.

Identify target users, including voters and administrators, and gather detailed requirements for each user role.

2. **Design and Prototyping:**

Create mockups or prototypes for the user interface, focusing on how users will interact with the system.

Design the architecture of the system, including how the authentication, voting, and result functionalities will be implemented in Python.

3. **Development:**

Backend Development: Implement the core functionality in Python, including user authentication, voting logic, and results generation.

User Interface Development: Develop a basic command-line or simple GUI for user interactions.

4. **Testing:**

Test the system for functionality, including user authentication, vote casting, and result generation.

Ensure that the system is secure and performs well under various conditions. Verify that all features work correctly and that the system is free of bugs.

5. **Deployment:**

Prepare the environment for deployment, including setting up any necessary servers or cloud services.

Deploy the application and ensure that it is accessible to users. Configure any required settings to make the system operational.

6. **Maintenance and Updates:**

Regularly update the system to address any issues that arise and improve functionality.

Monitor system performance and user feedback, and make necessary adjustments to keep the system secure and up-to-date.

3.10 Challenges and Solutions

1. **User Authentication Security:**

Challenge: Ensuring the security of user authentication to prevent unauthorized access and data breaches.

Solution: Implemented secure password hashing using SHA-256 and used robust encryption methods for data protection. Added multi-factor authentication for additional security.

2. **Handling Concurrent Votes:**

Challenge: Managing and processing votes in real-time while ensuring data integrity and preventing duplication.

Solution: Implemented mechanisms to handle concurrent vote submissions and used data validation to ensure accuracy. Utilized Python's threading or asynchronous features to manage real-time operations effectively.

3. **System Scalability:**

-Challenge: Ensuring the system could handle a large number of users and votes without performance degradation.

Solution: Designed the system with scalability in mind, optimizing code performance and using efficient data structures. Considered implementing load balancing and optimizing algorithms to handle high traffic.

4. **Error Handling and Debugging:**

Challenge: Identifying and fixing errors in a complex voting system where bugs could impact voting accuracy or system stability.

Solution: Conducted thorough testing and debugging, including unit tests and integration tests. Used logging and error tracking tools to monitor system behavior and quickly address issues.

3.11 Achievements and Learnings

During my internship program, I significantly enhanced my Python programming skills, particularly in areas like encryption and real-time processing. I gained a deep understanding of system design by developing a secure and scalable online voting system, focusing on robust architecture and functionality. This experience sharpened my problem-solving abilities, especially in tackling challenges related to user authentication and concurrent voting. I also learned important security best practices, including password hashing and data encryption, and acquired techniques for ensuring cross-browser compatibility and consistent performance. Handling real-time data efficiently and accurately was another key learning area. Additionally, I improved my testing and debugging skills, utilizing various tools to ensure system reliability. The internship also provided valuable insights into project management, teamwork, and communication. Lastly, I learned the importance of maintaining clean, organized code and adhering to best practices for code management.

4. Overall Internship Experience and Learning Outcomes

4.1 Professional Development

During my internship at Innovate Intern, where I worked on the SecureVote project, I experienced significant professional development that enhanced both my technical and soft skills. I applied my theoretical knowledge of Python to develop a robust online voting system, focusing on real-world applications such as user authentication, secure data handling, and real-time voting processing. The project exposed me to best practices in system design, security measures, and code optimization, which greatly improved my technical proficiency. Collaborating with industry professionals and peers provided valuable insights into effective coding techniques and system architecture. The hands-on experience also refined my problem-solving abilities as I tackled challenges related to concurrent voting and system scalability. Regular feedback sessions and the need to meet project deadlines honed my teamwork, communication, and time management skills. This internship not only prepared me for future technical challenges but also equipped me with a comprehensive understanding of developing secure and efficient software solutions.

4.2 Technical Skills Acquired

1. Python Programming:

During my internship, I developed a strong command of Python, focusing on implementing core functionalities for the SecureVote system. I wrote code for user authentication, vote processing, and result generation, which enhanced my understanding of Python's syntax and libraries. This experience involved using Python to manage user data securely, process real-time votes, and generate accurate voting results, all while adhering to best practices in coding and software development.

2. Encryption Techniques:

I gained hands-on experience with encryption methods, specifically using SHA-256 for hashing passwords and securing user data. This involved integrating encryption into the system to ensure that sensitive information remained protected from unauthorized access.

Understanding and applying these techniques improved my ability to handle data securely and reinforced the importance of incorporating robust security measures in software development.

3. Real-Time Data Handling:

Handling real-time data was a key aspect of the project. I learned to manage and process incoming votes efficiently, ensuring that the system could handle multiple submissions concurrently without compromising performance. This involved using Python's threading or asynchronous capabilities to maintain system responsiveness and accuracy, which is crucial for real-time applications.

4. System Design:

The project provided valuable insights into designing scalable and secure system architectures. I was involved in planning the overall structure of the voting system, including how to integrate various components securely and efficiently. This experience taught me to consider factors like scalability, reliability, and security when designing software systems, which are essential for building robust applications.

5. Testing and Debugging:

I enhanced my skills in testing and debugging by rigorously evaluating the SecureVote system. This included writing and executing test cases to identify and resolve bugs, ensuring that all features functioned as intended. I used debugging tools to troubleshoot issues and refine the code, which improved my ability to maintain system reliability and performance.

4.3 Soft Skills Developed

In addition to technical skills, I developed several soft skills, such as:

1. Problem-Solving: Working on the portfolio website enhanced problem-solving abilities by presenting various challenges, such as cross-browser compatibility and responsive design issues. The need to devise effective solutions for these challenges fostered a strong analytical mindset and the ability to tackle complex problems efficiently.

2. Attention to Detail: Developing a polished and functional website required meticulous attention to detail. Ensuring that every element was correctly styled, aligned, and interactive demanded a keen eye for detail, which improved the ability to identify and rectify minor issues before they became major problems.

3. Time Management: Managing the various phases of the project—design, development, testing, and deployment—required effective time management skills. Balancing these tasks within deadlines fostered the ability to prioritize tasks, allocate resources efficiently, and maintain a structured workflow.

4. Communication: Communicating effectively about project requirements, progress, and challenges was crucial throughout the development process. This involved articulating technical details clearly and collaborating with stakeholders or team members, which enhanced the ability to convey complex ideas in a comprehensible manner.

5. Adaptability: Adapting to new technologies and design principles was essential for the success of the project. Embracing new tools and techniques, such as responsive design frameworks and deployment platforms, improved the flexibility to adjust to evolving project needs and technological advancements.

6. Creativity: Designing a visually appealing and engaging website required creative thinking. The process of developing a unique design that effectively showcased the portfolio involved exploring various design concepts and finding innovative ways to present content, which nurtured creative problem-solving skills.

7. Self-Motivation: The independent nature of the project fostered self-motivation and discipline. Taking ownership of the entire development process, from initial planning to final deployment, strengthened the ability to stay motivated and focused on achieving project goals without external supervision.

These soft skills, developed through the project, are invaluable in both professional and personal contexts, contributing to a well-rounded approach to tackling challenges and achieving success in various endeavours.

4.4 Contributions to Innovate Intern:

At Innovate Intern, my primary contribution was in the development of the SecureVote online voting system. I played a pivotal role in designing and implementing core features, including user authentication, vote processing, and results generation using Python. One of my key responsibilities was to ensure the security of the system. I also focused on managing real-time data efficiently, optimizing the system to handle multiple concurrent votes without performance degradation. Extensive testing and debugging were crucial parts of my role; I conducted thorough tests to identify and resolve issues, ensuring the system's reliability and functionality. Additionally, I actively engaged in team collaborations, contributed to discussions, and incorporated feedback from mentors to refine the system. My efforts were instrumental in enhancing the system's robustness and security, providing me with valuable experience in software development and project management.

4.5 Responsibilities and Accomplishments

Responsibilities

1. **System Design and Implementation:**

I was responsible for designing the architecture of the SecureVote system and implementing its core functionalities. This involved creating user authentication mechanisms, vote processing workflows, and result generation features. My role required careful planning to ensure that all components worked together seamlessly and met the project's requirements.

2. **Security Integration:**

A key responsibility was to integrate robust security measures into the system. I implemented encryption techniques, to protect user data from unauthorized access. Ensuring the system's security was crucial for maintaining the integrity and confidentiality of the voting process.

3. **Real-Time Data Management:**

I managed the real-time handling of vote submissions, ensuring that the system could process multiple votes concurrently without performance issues. This involved optimizing data handling processes and ensuring timely and accurate vote processing to maintain system responsiveness.

4. **Testing and Debugging:**

I conducted extensive testing of the SecureVote system to identify and fix bugs. This included writing and executing test cases, using debugging tools, and refining code to ensure that all features functioned correctly. My efforts were focused on maintaining system reliability and addressing any issues that arose.

5. **Collaboration and Feedback:**

I actively participated in team meetings and collaborated with other interns and mentors. I contributed to discussions, shared progress updates, and incorporated feedback to improve the system. Effective communication and teamwork were essential for the successful development of the project.

Accomplishments

1. **Successful Implementation of Core Features:**

I successfully developed and implemented the core functionalities of the SecureVote system, including user authentication and vote processing. This accomplishment demonstrated my ability to translate project requirements into working software solutions and meet key milestones.

2. **Enhanced System Security:**

By integrating advanced encryption techniques, I significantly improved the system's security. The use of SHA-256 for password hashing ensured that user data was protected against unauthorized access, contributing to the overall robustness of the system.

3. **Efficient Real-Time Vote Processing:**

I optimized the system for handling real-time vote submissions, ensuring that it could manage multiple concurrent votes without performance issues. This accomplishment highlighted my ability to develop efficient solutions for real-time data management.

4. **Comprehensive Testing and Bug Fixing:**

My thorough testing and debugging efforts led to a reliable and stable system. Identifying and resolving bugs ensured that the SecureVote system operated smoothly and met quality standards, reflecting my commitment to delivering high-quality software.

5. **Effective Team Collaboration:**

I actively engaged with my team and mentors, contributing to collaborative efforts and incorporating feedback to enhance the project. This accomplishment underscored my ability to work effectively within a team and adapt to feedback, leading to a more refined and successful project outcome.

These responsibilities and accomplishments reflect a comprehensive approach to front-end development, demonstrating both technical expertise and project management skills.

4.6 Issue Resolution

I effectively resolved issues related to cross-browser compatibility, ensuring the blog functioned smoothly across different platforms. By identifying and addressing performance bottlenecks, I optimized the site's loading speed and responsiveness. I also fixed bugs in Python that impacted user interactions and thoroughly tested the site to ensure a seamless experience. My problem-solving approach enhanced the overall functionality and user satisfaction of the blog.

4.7 Staff Support and Mentorship

At Innovate Intern, the mentorship provided by Arvindan D Executive Director at Innovate Intern was instrumental in my professional growth. He offered invaluable technical guidance, helping me navigate complex development challenges. Also provided insightful,feedback on my work, ensuring that I adhered to industry standards and best practices. Their combined

expertise and support significantly enhanced my learning experience, enabling me to successfully complete my project with confidence and skill.

4.8 Challenges and Overcoming Them

During my internship at Innovate Intern, I faced several challenges, including mastering new technologies, ensuring cross-browser compatibility, and optimizing website performance. Overcoming these obstacles required persistence, continuous learning, and effective problem-solving. I sought guidance from my mentor, Arvindan D, who provided valuable insights and support. By testing extensively and refining my code, I successfully addressed these challenges, ultimately enhancing my technical skills and delivering a high-quality project.

4.9 Learning Outcomes

1. Enhanced Technical Proficiency: The project provided hands-on experience with Python, resulting in a deeper understanding of front-end development. This practical application of theoretical knowledge strengthened coding skills, particularly in creating responsive designs, implementing interactive features, and optimizing performance.

2. Improved Problem-Solving Abilities: The challenges encountered during the project, such as cross-browser compatibility and responsive design issues, honed problem-solving skills. Learning to troubleshoot and resolve these issues enhanced the ability to approach and solve complex problems effectively.

3. Mastery of Responsive Design: Through the implementation of responsive design principles, the project reinforced the importance of creating adaptable web layouts. This outcome led to a strong understanding of how to design python frameworks that provide a consistent and optimal user experience across various devices and screen sizes.

4. Advanced Project Management Skills: Managing the project from conception to deployment enhanced skills in project organization, time management, and task prioritization. These project management skills are crucial for efficiently handling future development projects and meeting deadlines.

7. Professional Presentation and Communication: The experience of presenting the python programming and communicating project details to mentors improved professional presentation skills. This outcome emphasized the importance of clear communication and effective feedback integration in achieving project goals.

8. Adaptability and Continuous Learning: The project highlighted the need for adaptability in learning new tools and techniques. This outcome encouraged a mindset of continuous learning and staying updated with industry trends and best practices.

5. Conclusion

5.1 Summary of Experience

During my internship at Innovate Intern, I worked on the SecureVote project, where I gained substantial experience in software development and system design. I was responsible for designing and implementing key features such as user authentication, vote processing, and results generation using Python. One of my major achievements was integrating robust security measures, to ensure data protection and system integrity. I also focused on managing real-time vote submissions efficiently, optimizing the system for performance and accuracy. Through extensive testing and debugging, I was able to ensure the system's reliability and stability. Active collaboration with my team and mentors allowed me to refine my skills and incorporate valuable feedback, enhancing the overall quality of the project. This internship provided me with hands-on experience in developing secure and scalable software solutions, improving my technical skills and preparing me for future professional challenges in the tech industry.

5.2 Future Prospects

The experience gained from my internship at Innovate Intern, especially through working on the SecureVote project, opens several promising future prospects in the tech industry. My skills in Python programming, system design, and real-time data management position me well for advanced roles in software development, such as software engineering and systems architecture. Additionally, the hands-on experience with encryption and data protection prepares me for a potential specialization in cybersecurity, focusing on safeguarding systems and ensuring data integrity. The expertise I've developed in handling real-time data can lead to opportunities in areas requiring real-time processing, such as financial trading platforms and live data analytics. Furthermore, the technical knowledge and problem-solving skills acquired could pave the way for advanced studies in fields like artificial intelligence or data science. Lastly, the experience of managing a complex project and collaborating with a team provides a strong foundation for future leadership and project management roles, enabling me to lead software development projects and drive technological innovations. Overall, this internship has equipped me with valuable skills and knowledge, setting the stage for a successful and dynamic career in technology.

5.3 Acknowledgements

I would like to express my sincere gratitude to Innovate Intern for providing me with the opportunity to work on the SecureVote project. Special thanks to my mentors and team members for their invaluable guidance, support, and feedback throughout the internship. Their expertise and encouragement significantly contributed to my growth and success in developing the online voting system. I am particularly grateful to Arvindan D, the Executive Director at Innovate Intern, whose leadership and vision were instrumental in shaping the project's direction and success. The hands-on experience and exposure to real-world challenges have greatly enhanced my technical skills and problem-solving abilities. The opportunity to work on such an impactful project has been a pivotal step in my career. I deeply appreciate the chance to contribute to the development of SecureVote and the unwavering support I received from everyone involved. Thank you for believing in my potential, providing me with this enriching experience, and helping me build a solid foundation for my future in technology.

5.4 General Evaluation

The internship at Innovate Intern, particularly through the SecureVote project, has been an exceptionally rewarding experience. The program provided a comprehensive exposure to real-world software development, allowing me to apply theoretical knowledge to practical challenges. I had the opportunity to work on critical aspects of the project, including user authentication, real-time data handling, and system security, which significantly broadened my technical skill set. The experience was further enriched by the guidance from experienced mentors and the collaborative environment fostered within the team. The project not only improved my Python programming abilities but also honed my skills in encryption, system design, and debugging. Regular feedback and hands-on problem-solving played a crucial role in my development, helping me refine my approach to software engineering. Additionally, the internship offered valuable insights into project management and effective teamwork. The leadership of Arvindan D, the Executive Director, was instrumental in steering the project and providing strategic direction. Overall, the internship has been a pivotal step in my career, equipping me with practical experience and a deeper understanding of the technology industry. It has prepared me well for future professional challenges and opportunities, reinforcing my commitment to pursuing a career in software development.

5.5 Skills Acquired and Future Development

During my internship at Innovate Intern, I acquired a range of valuable skills that have significantly shaped my professional growth. I developed a strong proficiency in Python, particularly in areas like user authentication, encryption, and real-time data processing, which are crucial for building robust software systems. My experience with implementing encryption and handling concurrent voting submissions provided a solid foundation in security practices and performance optimization. Additionally, I gained hands-on experience in system design, learning how to create scalable and secure architectures. The internship also honed my debugging skills through extensive testing and troubleshooting, enhancing my ability to ensure system reliability. Working closely with mentors and team members improved my collaboration and communication skills, while the feedback I received helped refine my problem-solving abilities. Looking ahead, these skills will be instrumental in pursuing advanced roles in software development, cybersecurity, or real-time systems. The technical expertise and project management experience gained during the internship have prepared me for future challenges, including opportunities to lead projects and innovate in the tech industry. Overall, the internship has laid a strong foundation for my career, and I am eager to build on this experience by exploring new technologies and continuing to grow as a software professional.

5.6 Career Impact

The internship at Innovate Intern, with a focus on the SecureVote project, has had a profound impact on my career trajectory. By working on real-world software development tasks, I was able to bridge the gap between theoretical knowledge and practical application, significantly enhancing my technical expertise. My involvement in designing and implementing key features of the online voting system provided me with hands-on experience in Python programming, system security, and real-time data management. These skills are highly transferable and have prepared me for advanced roles in software development and cybersecurity. The internship also exposed me to essential aspects of project management, including teamwork, communication, and problem-solving, which are crucial for career advancement. Working closely with experienced mentors and receiving constructive feedback improved my professional

competencies and boosted my confidence in tackling complex challenges. The experience has not only solidified my technical foundation but also expanded my understanding of industry practices and standards. This comprehensive exposure has positioned me well for future opportunities, such as leading development projects, specializing in cybersecurity, or pursuing further studies in advanced technology fields. Overall, the internship has been a pivotal step in shaping my career path, providing me with the skills, experience, and confidence needed to excel in the tech industry.

6.References

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